

Introduction

EVOLUTIONARY STRIPTease

Undress Rehearsal

Imagine a future in which three-dimensional holographic movies are shown, a future in which it is possible to recover DNA from fossils and “ghost” them—clone them, bring extinct beings back from the grave.

In such a future it would be possible to give a live presentation of what, in this book, remains purely a fantasy. The fantasy in question is that of a striptease. Not just any striptease, but an evolutionary striptease in which time present is put on rewind and our bodies are holographically unwound to reveal the salacious story of their sexual origins. In lieu of such a future—and because futures have the bad habit of never quite arriving—this book is set up as a kind of holographic theater, in which we show you evolution in reverse, from human to ape to reptile and beyond, exploring the evolution of men and women, sexual organs and sex roles, trusting commitments and erotic infidelities, love and lust.

Recent work by biologists has added much to our knowledge of sex. Drawing from philosophy and psychoanalysis as well as from rich scientific data this book tells a particular story of human sexuality. The “mystery dance” that unfolds on and

between these pages is an imaginary striptease, a thought experiment that spotlights a phantasm called the evolutionary stripper, a kind of genetic genie whose undressing takes us far back in evolutionary time. The stripper's act exposes the presumed sex lives and bodily appearances of our ancestors, human and prehuman. The dancer, for example, takes off an outer layer of civilized monogamy to reveal the wanton promiscuity of *Homo erectus*, fire-using hunters ancestral to *Homo sapiens*. By sexually climaxing, *Homo erectus* females helped choose the genetic composition of modern humans. Beneath this layer—that is, earlier in time—lie the characteristic sexual features of human beings. The backward-moving steps of the mystery dance display the evolution of human sexual features such as prominent breasts, large penises and heavy testicles, the hymen, and the absence in females of estrus—of regular periods of sexual desire accompanied by vaginal swelling and pubic coloration.

The animal body evolves not only in response to external pressures of the environment but also to the sexual tastes of the opposite sex and in response to competition with other members of one's own. This book's evolutionary striptease shows how the primeval kinks, the likes and dislikes of our sexually arousable ancestors, molded the human body as it evolved from its apish predecessors. The stripper also reveals the biological foundations of overwhelming emotional states, such as raging sexual jealousy and self-destructive desire.

Physiologically the most important steps of our sexual history were undertaken by our remote predecessors. They occurred in epochs long before the appearance of the animal known as "man." Such paleobiological happenings, rarely described, include the origins of the penis, the reptilian "hieroglyphic" brain, and protist sexual cannibalism. Most ancient of all our living predecessors are those bacterial inhabitants of the sweltering early Earth from which all life has evolved. They are not sexless: bacteria already avidly exchanged genes—and therefore engaged in the planet's first sex—long before the existence of either plants or animals. Today the ancient ability of bacteria to offer and accept parts of themselves as genetic

gifts in the formation of new, hybrid offspring is exploited in the laboratory, in biotechnology and genetic engineering.

The mystery dance is not exclusively the unveiling we call the evolutionary striptease; it is also the human sex act itself. From its obscure beginnings in the microcosm to the discovery of hormone birth control pills and mass production of rubber prophylactics, this mystery dance is our heritage. The slang word for coitus simultaneously means making love and an act of aggression. This duality is part of the secret of the mystery dance, an act that stands at the crossroads between the everyday and the transmundane, at that dusty fork where the vulgar meets the sacred, and reality hooks up with dream.

Shame

In 1584 women thought to be witches were being burned at the stake for committing "incestuous adulterie with spirits." A popular witch-hunting manual of the time asserted that the devil liked to assume the shape of a pretty wench and, in that form, seduce and make love to unsuspecting men. Keeping their semen the devil would then transmogrify into an incubus, a male sexual demon, and impregnate sleeping women as they dreamed. When Reginald Scot dared to publish that all this talk of men being "begotten without carnall copulation" was nonsense "speciallie to excuse and mainteine the knaveries and lecheries of idle priests and bawdie monkes; and to cover the shame of their lovers and concubines," King James of England ordered all copies of Scot's book burned.^{1*}

We still dwell in a sexual dark age, although scientific discoveries concerning the roles of hormones, widespread avail-

*The Notes and References section begins on p. 207. It includes commentary on epigraphs, peripheral remarks, and other potentially disruptive statements parenthetical to the text.

ability of contraceptives that prevent pregnancy and venereal disease, and a more open, information-oriented society all indicate that sexual ignorance may be abating. It is true that women are no longer burned at the stake. But the prejudice toward homosexuals; the stridency surrounding the questions of sex education, abortion, and pornography; the difficulties of distinguishing healthy attitudes from subtle abuse and exploitation—all these suggest that sexual enlightenment, if there is such a thing, has not yet arrived. In fact, the AIDS epidemic has had a dual effect: raising sexual consciousness and inspiring longer-term relationships on the one hand, and proliferating hypocritical moralizing and the scapegoating of gay men, Haitians, and intravenous drug users on the other.

Human sexuality resists the light of scientific inspection for several reasons. The widespread reticence and embarrassment that surrounds human sexuality result from its relation to those parts of the body that produce waste and over which our earliest job it was to exert control. "*Inter urinas et faeces nascimur*," wrote St. Augustine: "Between piss and shit are we born." (Putting it vulgarly emphasizes the inhibition we feel in the face of this fact.) Toddlers are taught to control their bowels and urine; this is their first focus, their prime directive. In consequence, they despise their excrement, which they hide and which, psychoanalysis tells us, they associate with adult genitals and the fantasy of anal birth. The child as he or she is socialized learns that the bathroom, the bedroom, and certain parts of him- or herself demand privacy, entail mysteries; the sphincters must be covered and controlled. Surrounded by secrecy, the genitals are ridiculed, and the realms of childbirth, bodily evacuation, and sexual intercourse are confused. Misconceptions multiply.

Part of the problem with the relationship between the sexes later in life stems from the fact that we are, most of us, not only born of but reared by women. The central figure in the infantile life of both boys and girls is female—a mother who is loved and hated, who provides and denies, who coddles and neglects the baby before it is able to distinguish its five senses, differentiate inside from outside, or recognize itself or other

human beings. As Dorothy Dinnerstein has shown in her book *The Mermaid and the Minotaur*, this early intense encounter with female humanity in the form of the mother has immense consequences: most of us in a dark corner of our mind continue to perceive woman from the prelogical vantage point of the infant, not as a human “she” but as an irrational “it,” an abstraction as dangerous and indomitable as “nature” or “the flesh.” Because of infantile experiences the body of the female remains magical, while the male tries to gain control over and access to that sort of body that was never sufficiently his own. What is male and human is defined in contrast to this early goddesslike power of the female. Dinnerstein suggests that men and women alike bear a lingering resentment against the woman because of the mother’s power over us as infants, and that the only way to alleviate the culturally reinforced fear and hatred of women is for men to share equally in the raising of children. Dinnerstein’s point is that the roots of sexual identity and feeling are buried in an infancy we have physically outgrown and “forgotten,” but that this infancy continues to haunt us, producing negative adult attitudes toward women; alternatively debased and exalted, worshipfully feared and resentfully controlled, she is, like the mermaid, never really considered to be fully human.²

Another reason for the darkness surrounding sex may be its biological “depth”: the sexual part of ourselves is so old, so deeply ingrained in our being as reproducing animal bodies that much of our sexual behavior is automatic, physiological, almost as deaf to our conscious wishes as the human heart, which beats regularly, naturally, as indeed it must if we are to live. In this sense sex is unconscious not because we have repressed its pain but because to be conscious of it would interfere with daily biological functioning. A final reason for the darkness surrounding sexuality is geographical: the West, unlike some Eastern countries, lacks a tradition of erotic spirituality. Western cultures have not integrated sexuality but marginalized it, kept it confined to the bedroom or on display as pornography, or used it to market wares. For the most part ours has been a tradition sanctioning sex within the family for

the sake of childbearing, while reluctantly admitting the necessity of discreet prostitution and the existence of covert homosexuality. We Westerners have no Kama Sutra, no Tantra, no sacred sexuality. We do not publicly celebrate lesbian lovers or the passionate embraces of youths. Even today homosexuality is often still, as Oscar Wilde said of it in the last century, “the love that dare not speak its name.”

Whatever its sociocultural expression, sexuality, intrinsic to human beings, reaches remotely back in time to a long line of nonhuman and inhuman ancestors. Beyond apes and past reptiles, amphibians, fish, and other vertebrates (animals with backbones), our sexuality descends from that of creatures who copulated and reproduced without anything resembling human form or behavior; this is why sex is no mere human phenomenon. We are indeed like mermaids and mermen in that we remain in touch with our former selves, a fleshy pastiche not just of fish but of thousands of ancestral lives. And if biology tells us that our body-minds embed their own ancient history, so too, psychoanalysis tells us that adult life is never completely released from the experiences of infancy and childhood. Sexual passions, fervent and confused, continue from a past we can ignore but not escape.

We begin to understand the sexual being of adult humans by tracing its hardly imaginable distant history. Fewer than one *million days* have passed since the birth of Christ; this book tells tales of bacterial beings living three *billion years* ago. The molecular, intracellular activities of these beings still exert a crucial influence on our sexuality. And so do the forms of our more familiar relatives. We jump forward prior to, shortly after, and during the appearance—according to the fossil record, four million years ago—of humanlike animals. Bones of australopithecines, *Australopithecus afarensis*, were unearthed near Hadar, Ethiopia. The males of some of these ancestors were nearly twice as large as the females: on our evolutionary lineage, physical domination by males and feminine submission may be ancient phenomena.

It is folly to ascribe human sexual behavior solely to biology, to immutable genes. Social Darwinists, politicians, and even

academics have used the idea of innate biological differences to justify sexism and racism, causing feminists and liberals to dismiss biology as “determinist.” Biology, they claim, is “biologicistic”; by portraying the contingent historical knowledge of our own culture as the immutable truths of all nature it provides the perfect excuse for maintaining an oppressive social status quo. But although acutely aware of the potential for erudite bigotry, we as a writer and a scientist nonetheless embrace evolutionary biology. Branding women as “inferior,” some African races as “less evolved,” or using such claims to smugly pardon social inequalities as “natural” are all despicable; such practices malign not only the scientist’s case but the case for science. Similarly, the claim that humans are so genetically “programmed” that “instincts” are always impossible to transcend is ridiculous. But to reject evolutionary biology because of the excesses of its invokers is like throwing away a pearl because it is surrounded by a strong-smelling, seaweed-entangled oyster.

There is a strong case for the deep influence of our evolutionary history. The genetic potential for development is inherited by all animals, including, of course, people. Many behaviors are clearly genetic. All children, for example, sleep, even in the womb. Intrauterine studies show that the unborn have REM—rapid eye movements—just as dreamers do. Sleep is not a learned or cultural trait: it is genetic (inherited) and congenital (present at birth). All babies cry and reach for the breast to suck. Children, on the average and compared with adults, have a superior ability to learn a new language. Although neither English nor Chinese are “built in” to the genes, it is plausible to argue that the ability to babble and then to speak is inherited and biologically timed such that children learn to become members of their community, full-fledged linguistic, human beings.

Nature and culture are far from clear-cut, independent categories; rather, they intertwine. Just as our bodies grow unconsciously within the womb and beyond the womb under the choreographic powers of the genes, just as we breathe from birth without the benefit of advice from doctor or “culture,”

so aspects of our sexual behavior are unlearned and inborn; they are “bred in the bone.” And these are aspects, we believe, that are as embedded as they are unconscious, wielding great power over our lives as sexual beings. Far from dismissing those features as socially taught habits that can be unlearned, the best way to begin to outgrow them—if outgrow them we can—is to realize just how deeply—biologically—enmeshed in our being they are.

To what extent we can overcome the influence of the past is a problem in itself. An early reader of this book, Madeline Sunley, puts it eloquently: “One of my favorite questions about the future of our species is what I’ve come to think of as the horrified spectator question. Can we change destructive behaviors once we are aware of them? Or are we, at least at this stage of human evolution, trapped in a state of cognitive dissonance in which, powerless to change, we watch ourselves behave in ways we do not condone? This is one of the questions which lie at the very heart of a book like *Mystery Dance*.”

Meaning and Being

It is easy to discuss human sexuality with clinical detachment, protected by a cloak of authorial objectivity or scientific expertise. But to talk of the mystery or meaning of sex is far more risky. Blanket statements verging on nonsense open a roomlike space in which sex can be seen for what it is. This book with its striptease invites a sense of voyeurism. In examining the human sexual experience, the reader sacrifices the security of “objective” knowledge to discover—and perhaps feel—more than he or she wants to know. Reading, you slide between the covers of a bedlike book as you confront these ink-stained sheets. Open eyes shift from side to side, just as they do during REM sleep. Indeed, in a sense this whole book is a kind of science “faction,” a public, waking dream.

The French philosopher Jacques Derrida is the source of the intellectual ferment over “deconstruction.” It is dangerous

to define deconstruction, on the one hand because it has become such a loaded term, and on the other because its business has largely to do with subverting the conceptual, metaphysical categories policed by definitions. Deconstruction destabilizes texts—and ultimately the very notion of self-identity—by grafting them onto each other, unravelling weak arguments, and showing that authors often betray their own intentions. But Derrida has exploited the fact that the French word *lit* means both bed and read. Derrida playfully evokes this ever-present sexual underside of meaning even in the loftiest, most serious writings. For example, the English verb *mean* shares roots with *moan*, and we all know how quickly, once sought, sexual connotations surface. Even terms designating fields concerned with the study of meaning—such as *semantics* and *semiotics*—bear an evocative resemblance to the sexual word *semen*.³ Such double entendres may seem distracting to the scientist but they are unavoidable. Moreover, they point beyond mere clinical sex research and the “facts” of biology to a compelling or compulsive study of the sexual depths of the human psyche.

According to the nineteenth century philosopher Friedrich Nietzsche, when women in ancient Greece tore apart and ate live chickens as they paid homage to Dionysus such homage to the god of wine evoked “above all the frenzy of sexual excitement, this most ancient and original form of frenzy”.⁴ The Dionysian pleasures of the feast, of *ecstasy* (literally: standing out of oneself), of conversation as psychosexual intercourse, of melding with or disappearing into the body of the other, of almost dying (a French expression for orgasm, *petit mort*, means “little death”) . . . these are phenomenal experiences of sex that precede any high-handed scientific evaluation of human sexuality. A candid exploration of human sex leads to problems of meaning and being every bit as important and in some sense prior to the specific issue of the evolution of sex. Martin Heidegger, perhaps the most influential philosopher of the twentieth century, considers what has been translated as *Dasein*—“being there” or “human being”—along with *Sein*, “Being”—to be more primordial than any

science such as psychology or biology. The study of being is called ontology. Some philosophers distinguish between the ontic—the realm of individual beings—and the ontological—“Being” in general. For Heidegger, both come before specific “derivative” sciences such as biology, psychology, and more broadly speaking, “technology,” which although not a science like the others, is crucial for philosophy, according to Heidegger, since it defines the metaphysical character of our age. In addition to everything else, sexuality is ontic, part of our existence as living beings. And it may even be ontological, part of Being; to be alive is to experience a fundamental solitude, a loneliness that may be alleviated by the other, the lover, but never completely. This fundamental isolation of individuals from the universe Heidegger calls the “ontological difference.” Sexuality comes into play because, however imperfectly, it provides a ready means of attempting to bridge this difference. Similarly, the mystical Eastern traditions suggest that individual personalities are only masks, and so, when lovers unite in sex, they may call forth a distinct state, in which the “I,” the self, is seen to be an irresistible illusion.

The Stripper

The stripper in this book is androgynous. As (s)he takes off his/her clothes, we will be treated to views of our ancestors in earlier times. The stripper also is endowed with multiple identities. Sometimes (s)he will appear as a single person, a man or woman, sometimes as a mating couple, an animal, or a collection of microorganisms. The stripper, like the self in Eastern philosophy, has no discrete “I” but appears in all these guises. Each guise represents an evolutionary ancestor. Our working assumption is that virtually each of humanity’s ancestors has left its mark, has helped mold human flesh and human nature in its ambiguity and complexity. Peeling off one layer presents us with views of the early human species, *Homo erectus* and prehuman lovers of the genus *Australo-*

pithecus (of which the famed “Lucy” skeleton is an example). We watch the changing stripper to find why apish females lost their body hair, pubic coloration, and developed permanently pendulous breasts. We learn that the woman’s body itself is a form of practical feminism, deceiving males all too desirous of inseminating fertile females. And we find that the transfer from ape to human gave females greater control over their bodies, countering with cunning the superior physical strength of most males. But the ape in us, as important as it is, is only one guise, or disguise, in a striptease that brings us all the way back to our bacterial origins.

Peeling off another layer, the stripper exposes beneath the warm mammalian fur and the reassuringly familiar mammalian psyche, a cool level of “reptileness,” including the “reptilian” brain—an ancient part of human anatomy shared not only with apes but with all mammals and reptiles. The R-complex, as the reptilian remnant of the human brain is sometimes called, remains. Overriding rational consciousness, this sex-attuned control center apparently undermines well-intentioned human beings, provoking reactions from jealous rage to impetuous abandonment and sudden passion. Modern reptiles lack keen hearing and an informative sense of smell; they *see* a patterned environment. Processing visual cues more with the retinas of their eyes than the cortex of their brains, reptiles communicate quickly, automatically, and by largely inborn means. Permanently “dumb,” reptiles cannot learn to speak. Much of what we consider inhuman in ourselves may depend on an underlying reptilian sensibility, an archaic substratum that may suddenly and powerfully surface. Reptiles are dreamless, amoral slitherers who will kill on the spur of the moment but rarely form emotional attachments, even to their own egg-emerging offspring. It even seems that the mammalian ability to construct events into a chronological order occurred during the transition from the reptilian to the mammalian brain. Paradoxically, the sense of passing time itself evolved in these ancestors of ours. Dinosaurs and serpents—and our reptilian ancestors—cannot grasp stories. They appear to live in a permanent present, the waking reptilian equivalent

Geological Time Scale and Sexual Ancestors*

EON BEGAN	EONS	ERAS
0†	Phanerozoic	Cenozoic 66–0
		Mesozoic 245–66
		Paleozoic 580–245
580		
2,500	Proterozoic	PrePhanerozoic
3,900	Archean	
4,500	Hadean	

*Not to scale and simplified.

†Dates in millions of years ago.

PERIODS		EPOCHS		ANCESTORS
Neogene	26-0	Recent (Holocene)	0.1-now	Ape-people: <i>Homo sapiens</i> <i>Homo erectus</i> <i>Australopithecus</i>
Paleogene	66-26	Pleistocene	2-0.1	
		Pleiocene	7-2	
		Miocene	26-7	
		Oligocene	38-26	
		Eocene	54-38	
		Paleocene	66-54	First primates
Cretaceous	138-66			Age of reptiles
Jurassic	195-138			
Triassic	245-195			
Permian	290-245			Thecodont reptiles (dinosaurs and bird ancestors)
Carboniferous	345-290			
Devonian	400-345			
Silurian	440-400			Synapsids (mammallike reptiles)
Ordovician	500-440			
Cambrian	580-500			
				Protist mates
				Bacterial symbionts

to our nighttime dreamworld. But they can be moved by events and react quickly to a sign they perceive as a mate or threat.

Mammals and dinosaurs both evolved from an earlier group of so-called stem reptiles. Reconstructions from fossil bones reveal extinct ancestors, dog-toothed lizardlike beings and reptilian “weasels.” Fossils of such mammallike reptiles abound on every continent but Antarctica. One paleontologist calculates that some 800 billion skeletons of mammallike reptiles exist in the Kaaroo beds of South Africa alone.⁵ Evolving about 250 million years ago, populations of these creatures diversified prodigiously during the Permian and Triassic geological time periods (see Table).

The fossil record can be interpreted to mean that when the forerunners of the giant dinosaurs evolved—swift and vicious thecodonts—only a few mammallike reptiles survived. Unable to defend themselves against their increasingly brawny and ferocious cousins, the earliest mammals took to nightlife, hiding in the dark, in trees, and relocating to cooler climes, where they were free from molestation. These remote human ancestors, the mammallike reptiles or synapsids, were physically weaker than the thecodonts. Many fell prey to the fierce predators, but expanded sensory modalities—particularly the sense of hearing—evolved in those of our reptilian ancestors that survived. Cries, scratches, howls, and scurryings in their nocturnal dwellings warned skittish synapsids of the approach of aggressors and the retreat of potential prey. But in addition to these increased sensibilities, our four-legged forerunners retained the original deadly psychology of their reptilian heritage, their narrow focus on brute survival in the tropics: to kill, to avoid being eaten alive, and to copulate—these were the none-too-subtle means that had always ensured their reproduction. The foray into our reptilian ancestry, into the strangely familiar psychology of reptiles, brings us into contact with psychology. And its most radical offshoot, psychoanalysis.

Freud’s French follower, Jacques Lacan, insists upon the absolute central importance of the phallus as a symbol or signifier—not so much for the penis as for what is missing—a lack-in-being (*manque-à-être*) or castration at the heart of

all would-be communication. The phallus is an erotic arrow pointing beyond the confines of evolutionary biology into the dark continent of psychoanalysis. It leads us from a discussion of the evolution of the penis back again to familiar territory whence humanity came. Although we are by no means completely convinced by them, we stop to discuss theories of the phallic signifier, the mirror stage, and the infant's magical psyche as the evolutionary dance continues and the striptease recedes further into the womb of time.

The large average size of the human penis (five to six inches, versus three for chimps, and one-half that for gorillas) may have been, evolutionists speculate, to frighten other males. Or to attract females. Or to enhance their pleasure. Perhaps the best hypothesis is that the longer penis delivers sperm more closely to the eggs: today's biologists claim that for females who mated with several males, the male with the longest penis delivered his sperm more safely. The first penislike organs probably arose in fish or amphibians whose ancestors had reproduced solely by external fertilization, by spawning outside bodies, in the warm water of lakes or the frothy ocean. The warmth and wetness of the human sex act recalls the breeding grounds—water—of those amphibians and fish ancestral to reptiles. As the mystery dance continues, our lizardlike fore-runners disappear. These were disseminators of sperm who competed for females without benefit even of a penis and shook the muddy water off their smooth and shiny bodies before slinking on.

Before our amphibious ancestry and beneath the stripper's amphibious dress lie the soft-bodied animals sometimes called Ediacaran fauna after the place where they were first found. But these strange gelatinous beings are rarely preserved in the rock record and may not contain any of our direct antecedents. And if quivering jellyfishlike animals are difficult to find, the colonial microorganisms that preceded them are still more elusive.

Wet and slippery, sex does not fossilize well. Unlike trilobite animals languishing on ancient seashores, insects trapped in sticky plant resin before it becomes amber, or ape lovers drag-

ging their feet and thereby leaving footprints in a romantic stroll through drying mud, the microscopic events at the heart of sex are rarely preserved in the rock record. Life is thought to have begun as oily molecules formed in tide pools of warm shallow seas; the first life, bacterial slime, spread as a kind of briny, colorful mucus—the slippery “microbial mats” or hard domes called stromatolites that still today in remote places such as Western Australia creep inexorably toward the sunlit surf. In both a physical and metaphysical sense, human sexual activity returns the body to the softness of its marine origins, to a time when life had not yet hardened, protected, and extended itself by incorporating durable substances such as lignin, shell, and bone. The engorged, the damp and brackish genitals engender the warm environment of ancient genes. They re-present a past present when single-celled beings looking much like sperm and ova existed but the human bodies that now produce them did not. It is among these cells—protists—that look so much like free-living sperm or egg cells that some of our most remote ancestors are to be found.

The mysterious and highly varied sex lives of protists hold secrets and clues for a comprehensive understanding of human sex. Yet even before the protists, the bacterial progenitors of all life lived on Earth. Some of the oldest unmetamorphosed rocks on Earth bear witness to ancient bacteria caught in the act of dividing. In Swaziland in southeast Africa are sedimentary rocks that, cut into thin strips and viewed under the high-power light microscope, reveal fossilized fission, the most simple means of reproduction known. But such bacterial fission is virtually the opposite of sex, of fertilization, of cell fusion: it is reproduction in the absence of sex. No fossil yet found preserves protist cell division, the process called mitosis that makes two perfect copies of the parent cell. And certainly no meiosis—truly sexual cell division making sperm or eggs with only half the parental number of chromosomes—has ever been found preserved in the fossil record. Yet fossil flower parts and petrified egg shells attest to the ancient history of meiotic sex. It is because of this second cell division process, meiosis, that animals like us must seek out and join up with the opposite

sex to make a new fused, or fertilized, cell, a cell that squares the circle, makes the half a whole again. Meiosis has long been a cellular imperative: genes that survived into the next generation depended on the meiotic production of egg cells and sperm and their fusing within the female's fallopian tubes.

Since meiotic sexuality is present in some protists, absent in others, and represented in a sort of half-hewn or midway state in still others, it seems likely that the kind of sexuality and cell fusion that brings man-sperm and woman-egg together began in watery protists, most of which are invisible to the unaided eye. The question is, how did protists—these microbes, more complex than the bacteria but less complex than the first animals—ever hit upon the trick of fertilization: doubling up their nuclei, chromosomes, and genes once each generation? And what is the point of this doubling, when the doubled cell only splits up again by meiosis to form sperm and egg? "One routinely recites the horrors of sex, metabolic, venereal, and behavioral, with the inevitable punch line, 'And for all your troubles your children carry only half your genes,' " writes University of Virginia biologist Robert Kretsinger.⁶ Although Kretsinger suggests this paradox is used mainly to titillate students despondent about grueling exams, evolutionary biologists have, in fact, debated it at great length. The potential answers are complex but often revolve around the idea that meiotic sexuality must confer some great benefit—for example, that it must somehow "speed up" evolution, or have some other function that keeps it from disappearing. Scientists have reached no consensus. The actions in the evolutionary striptease, however, underscore our idea that meiotic sex, in and of itself, may have no clear evolutionary reason or purpose. Our tiny ancestors simply got caught in a ritual dance, now beyond mere habit, of cannibalism, cellular fusion, and partial regurgitation. Without immune systems, it was possible for our microbial ancestors to swallow each other whole, digest only part of the meal, and extrude the living rest. Like many a distasteful survival story it sounds disgusting, but this first meiotic sex worked: those beings that took the bitter pill of protist cannibalism persisted; they lived to pass on their genes.

Microbial sex—meiosis and fertilization—has cast a long shadow over the existence of animals. While all-female species of lizards and rotifers reproduce parthenogenetically—without benefit of males ever—even they retain unmistakable traces of the cellular dance of meiosis. The peculiar sex lives of protists linger on, and the protists at least go through the motions of the billion-year-old events crucial to their reproduction, if not feverishly and faithfully replaying them with no loss of original ardor.

No person today dispenses with the return every generation to “germ” cells. The eggs that menstruating women shed and the sperm that men ejaculate each have only a single set of chromosomes. Until their fateful meeting inside the woman—an old saying states that life begins and ends at the back walls of the vagina—the sperm and ovum each still possess only one single set of chromosomes. All people—embryos, fetuses, babies, children, and adults—absolutely require two complete sets of chromosomes to live. More than any body part, these chromosomally stripped cells, squirming sperm and sticky ova, resemble our free-living protist ancestors. Ensuring their doubleness, the microbeasts became stuck in the original sexual relationship, a slippery routine of merging and emerging, smothering and removal, bondage and escape. They live on deep inside us.

But sexuality started much earlier than the cannibalistic gorging cells with nuclei that engulfed each other during times of drought and starvation. Sex began on a planet whose only inhabitants were the colorful, promiscuous bacteria, tiny, colonial beings incessantly exchanging their genes. Sex began during the Archean eon on a planet with an atmosphere very different from that of today. This planet was zapped by lightning and scorched relentlessly by ultraviolet radiation from the sun. As the evolutionary stripper will show us, bacterial promiscuity most likely arose from chemical mechanisms of DNA repair that evolved in cells damaged by solar radiation.

Bacterial sexuality differs fundamentally from the sexuality of organisms with nuclei (protists, plants, fungi, and animals), in that it occurs independently of reproduction, crosses “spe-

cies" barriers, and involves, in principle, the sexual sharing of genes by bacteria all over the world. Indeed, Canadian bacteriologist Sorin Sonea points out that bacteria, since they are able to trade genes freely across would-be species barriers, are not really assignable to species at all. In nature, bacteria with their uncounted numbers and metabolic types form a global "superorganism" whose bodily contours are those of the biosphere itself. This superorganism is unequivocally "sexual": it is continually trading genes among its myriad parts. It literally has sex with itself. Although sex in bacteria does not lead to single offspring with two parents, as it does in reproducing mammals, sex in the microcosm leads to genetic novelty. The global community of bacteria, which accounts for the transport of carbon, nitrogen, and all other biogeochemical cycles of the biosphere, is nearly four billion years old. Such age is almost inconceivable, verging on immortality. To this age-old superorganism the entire odyssey of humanity from shaggy erect ape to computerized corporation would appear no more substantial than the passing fancy of a butterfly. Indeed, compared to the solid tenure on this Earth of the bacteria-run biospheric superorganism, the stay of *Homo sapiens* is a mere blip.

It might seem that the microbial phase is the stripper's last, when (s)he loses everything and the dance is done. But there is perhaps a still deeper phase, the metaphysical plane of pure phenomena, continuous appearances. The evolutionary stripper is a curious creature: the G-string is not a thin cloth decorated with tassels but rather a word, a letter, a musical symbol for ultimate nakedness. Paradoxically, when the G-string is removed—to the accompaniment of strange vibratory music, consisting in part of a silent triangle and a gentle crash of cymbals—the nakedness itself is gone. S(he) stands before us as fully dressed as ever before.

At first glance, with the genetic exchange in bacteria, the most elemental living beings, it appears that we have reached the end of the evolutionary striptease, reduced sexuality to its essence. But no: there is always more. Not only the nature

but even the possibility of any final revelation is in doubt. The whole masquerade of the evolutionary striptease depends throughout upon a type of information completely different from molecular biology's genes but no less essential: phonemes, graphemes—words. We encounter our sexual ancestors along the slippery slope of signs and signifiers, via the medium of language. The use of signs of any kind necessarily obscures; words represent or replace signified things in absentia; they are little black masks. We postpone reality to discuss it; without this postponement, this instantaneous replacement of our sexual ancestors, or things in general, by their signs there could be no possibility of language, of signification at all. Language is dead material: a burnt stick, a sliver of bark, substitution for the living presence. But it is not only the little black marks of the alphabet that stand in the stead of real things, or at least of real sounds: the whole world is at a remove from itself. There is no simple purity of presence.

Jacques Derrida claims the idea that there is any presence to start with, that there is, as Gertrude Stein might say, any "there there," is itself in question. The idea of presence, a metaphysical construct, is not limited to writing, but is applicable to writing in the wider sense given it by Derrida. "There is no outside-text," writes Derrida.⁷ This means: nothing simple existed freely at the origin that unambiguously gave rise to complexity, obscurity, and impurity. Complexity, obscurity, and veiling are always already with us. Such thoughts are important for our idea of knowledge through discovery because they suggest that despite the continuous unveiling the evolutionary striptease conceals as much as it reveals. Truth cannot return to a revealing nakedness it never had.

Jean-Jacques Rousseau longed for the innocence of his unspoiled youth, the fresh smells of the countryside, and the dancing of children. Derrida describing this in *Of Grammatology* contrasts Rousseau's idea of an original innocence and plenitude with the feelings of loss—of loss of "presence"—to which Rousseau guiltily confesses in descriptions of "conjuring forth absent beauties" while masturbating. Rousseau, as Der-

rida shows, yearns in vain. At the origin the presence is not a presence; it is already the memory of a presence. The evolutionary dancer, the exotic chronicler of our sexual past slides out of personhood again and again in the frenzied attempt to show us the truth of our origins. But (s)he cannot. Instead of the past being disclosed, we confront an idea of the past in the present. (S)he is fickle. Now (s)he shows up as the scientific truth of our past, now as a naked metaphor: a paper dress patterned with these very words.

We are almost ready to enter the imaginary theater, dim the lights, and watch the holographic machinations of the evolutionary stripper as (s)he begins undressing. But first a brief note on time.

Time

Most people think of time simply as a passing stream of “nows” in which what has passed away is in the past, whereas what is future has still to come. We speak of “water under the bridge” and the “flow” of time. But this common image of time as a stream may be largely due to language. Indeed, some linguists believe that speaking different languages coincides not merely with different words but with distinct ways of thinking, of ordering and perceiving the world. If true, then language directs our view of time. For example, occasionally we speak of what’s “around the corner” or “over the horizon”—seeing our future as a place blocked to vision, but generally the language leads us, as English-speakers, to think that our past is *behind* us. A different “past,” by contrast, has been attributed to native speakers of the Navajo language. History, in the spatial metaphor of the Navajo language, is *in front of you*.⁸ This makes sense: we can see where we have been; it is the future, not the past that remains opaque, impossible to see.

The Navajos back into the future with the panorama of the past before them—as if life were a train ride in which the passenger faces the rear. Keeping in mind this “new” Navajo view of time, we turn our mind’s eye to the stage, where the past is clearly laid out ahead of us—far more clearly than the future at our backs.

PHALLIC PSYCHE

Like the Tower of Babel, they knew what they were after.

—PATTI SMITH

Sexuality is the vanishing point of meaning.

—JACQUELINE ROSE

The evolutionary stripper turns to reveal an erect penis. "She" becomes a "he." The phallus, conspicuous in many ancestral bodies, is now gone. In its place float two fish holograms, swimming dreamily through the air. The fish are preceded by shimmering green amphibians and mercurial pools of water, reflecting the spot of light that hovers above the stage.

The evolutionary dancer has holographically undressed men and women to display the spectacle of our mammalian and reptilian innards. But in the darkened theater of motley flesh and Navajo time we have never been sure which figures portray our real prehistory and which remain mere vivid fantasies. Moreover, as we move beyond a hundred million years ago into the realm of reptiles, it becomes increasingly unclear what part of the dancing hologram is meant to represent body and what part mind. Now, as if to emphasize this confusion, the evolutionary

stripper devolves into an amphibious morass in the shape of a lithe woman, then shatters and reforms into a gelatinous mass of eggs, glittering like a million pearls. Dissolving, melding together, the glittering caviar of amphibians becomes a smooth silvery mirror. The spectators search the shining, laser-made crystal for the clue to some secret meaning, but they find nothing but the spectacle of their own inquisitive faces. The mirror melts into streams of mercury that collect into a pool on the floor of the stage. As the pool of quicksilver shrinks, two fish with modified limbs float over the stage, then land on it, catching the air audibly as they prance rhythmically, in stylized fashion, trying to see beyond the edge of the spotlight's comically moving beam.

Psychoanalysis

The phallic psyche: in this chapter we will gingerly take a step off the relatively solid ground of science into the more speculative domain of psychoanalysis. Here we will become acquainted with some of the basic ideas of contemporary psychoanalysis. In this chapter we learn not only of the evolution of the physical penis but of the phallus as a source of symbols and meaning in the “timeless” unconscious. Since the phallus is the most obvious organ of sexual reproduction, it becomes a basis for childhood fantasies, and a focus for the magical thinking studied by psychoanalysis. We will briefly look at this magical thinking—the thinking that reveals itself in jokes and dreams and Freudian “slips”—and see how it seems to work by magic words and illusory mirrors. Then we return to the more empirical description of the prehistory of sexuality.

Is psychoanalysis just literature and poetry couched in the presentable form of serious scientific discourse? No, not “just.” Psychoanalysis, as widely understood in Great Britain and the United States, is a form of therapeutic expertise; but it is not only a means of healing the sick. Psychoanalysis is alive and

well, now thriving in adulterated forms under names such as “deconstruction” and “literary criticism.” The resurgence of the works of Freud, mainly due to French intellectuals, is not necessarily welcomed or even well known to English and American readers. Attitudes toward psychoanalysis are well illustrated by an anecdote told by the Canadian psychoanalyst Francois Peraldi. Peraldi describes a transatlantic encounter, a meeting of the minds between several French and several American psychoanalysts. Papers written by the participants had been distributed before the meeting, but a deep hush fell over the group when asked to begin. Finally, after several moments of silence, an American psychoanalyst spoke: “Well,” he said, “we have read all your papers but we are sorry to say it is not psychoanalysis. It is literature.” “We, too, have read your papers,” replied a French psychoanalyst, “and it also is not psychoanalysis. It is medicine.” Although psychoanalysis is differently interpreted, we believe that in it the spirit of scientific inquiry remains absolutely firm. Biology is a main pillar in the edifice of psychoanalytic thinking begun by Freud. But psychoanalysis goes beyond biology. Describing it Freud said psychoanalysis goes on and on; never coming to closure, stopping without ending, it demands patience and fortitude from those who would really undergo it.

Psychoanalysis occurs not only between analyst and analysand within the privacy and comfort of the analyst’s office. Confirmation of an expensive and exclusive appointment in advance, the supine patient who “free-associates” from the reclining comfort of the couch, the private office of the stone-faced analyst jotting notes—these are, in a sense, mere embellishments, stereotypes of psychoanalysis, Freudian clichés. Authentic psychoanalysis—the exploratory breakdown of the “soul” or *psyche*—can also take place between authors and readers; one of the ways it does so is by a stream of consciousness—a presence of mind—followed by the abdication of authority. With the breakdown of the psyche, of authoritative answers, it becomes unclear who is speaking to whom. A book, too, is a psychoanalytical tool; it generates communication by provoking thought of oneself with oneself. Indeed, the words

on a page of a book do not respond—they are irresponsible—like the attentive but taciturn psychoanalyst. The payment of money, Freud insisted, alters the nature of personal and emotional ties, underscoring the professional detachment of the analyst. And yet expenditure is important not only to the success of psychoanalysis but to the acquisition of a book such as this: so p(l)ay attention. Psychoanalysis was dubbed by one of its first patients as “the talking cure.” Might it not also be, in keeping with these parallels, a “writing cure”?

The Phallus in Magical Thinking

For psychoanalysis the phallus, the penis as symbol, is “the missing something”—that ever-elusive satisfier that permanently sustains desire precisely because it is never really obtained. Examining its evolutionary origin, we see how, in the ancestors to reptiles, the penis appeared in evolutionary time. Piercing the superficial genitals and indeed the entirety of the body we will continue to search for a missing something in the microbial layers that form the “heart” of sex as it is physiologically or biologically construed. And yet psychoanalysis warns us that such a search may be interminable, revealing no final key or golden treasure at the end of the researcher’s quest. To understand the absence of finality, we make a detour along the none-to-pithy path of psychoanalysis.

Evolutionary biologists, we have learned, insist that the penis because of its crucial role in reproduction is of greater importance to survival than is the clitoris. But does this high survival value afforded the penis support Freud’s view of “the” unconscious mind? Psychoanalysis, like evolutionary biology, emphasizes the primary importance of childhood views of the penis—or rather, the phallus, the penis as symbol. Phallic symbols for Freud are virtually omnipresent, especially in their absence. Indeed, the young girl’s view of male genitalia leads her to presume that she is inadequate, castrated, missing something of vital importance. For Freud this castration takes on

special significance. Freud called the female's desire to replace the lost phallus "*penisneid*." Translated, this is "penis envy," but that appellation is too literal. The phallic symbol is not simply an object that can transpose or transmute into any number of cylindrical forms. When Freud was asked if his cigar were a phallic symbol he commented, "Sometimes a cigar is just a cigar."

For Jacques Lacan desire is always desire of the other and this means, often enough, desire of the mother, the first love object; since the mother lacks the phallus, the child, whatever its sex, wants initially nothing less than to *be* the phallus for the mother. The child realizes that he or she comes out of the place where the father with his penis goes in. The reason the child wants to be the phallus is simple. He or she thus magically associates him- or herself with the male organ. Having the phallus, as boys do, is already a form of castration, since it does not satisfy the desire of the mother. Thinking magically, children identify themselves with what goes in or comes out of the mother's body. At the same time they realize the inadequacy of this identification. Psychoanalysis is largely an attempt to explore the magical thinking of the child that lingers on in the adult.

The male corollary to *penisneid* is the "castration complex": the little boy revels in his penis, he adores it but feels its inadequacy and fears he will lose it, as (he believes) his mother did. Desiring to be one with the mother, the threat of penis loss is his just punishment for wishing his father away. But since the boy's penis only approximates and is not the real organ desired by the mother, Lacan argues, not only girls but "everyone is castrated."¹ Psychological health requires accepting the reality of these magical thoughts of castration, not in the sense of acknowledging the absence of genitals but in the sense of being aware of a permanent inadequacy in the realm of desire—especially on the level of language. Acceptance of "castration" means coming to grips with the repressed childhood thoughts of permanent inadequacy and loss.

The psychoanalytic phallus thus need not be a penis. Since it means or points to something other than itself, it is not self-

identical. In the unconscious mind, as explicated by psychoanalysis, the phallus is always on the verge of something else. The “surreal” dream logic of the phallus can be appreciated by looking at René Magritte’s 1942 painting *La Mer* (The ocean) of a reclining nude male whose penis is a woman. Freud claimed that the shoe of a shoe-fetishist signifies the “woman’s (mother’s) phallus in which the little boy once believed and does not wish to forego.”² Beyond Freud, Lacan wrote, “We know that the unconscious castration complex has the function of a knot.” Mischievously, the word Lacan uses for “knot,” *noeud*, is a well-known French slang term for penis.³ Both Freud and Lacan use their writings not just to describe but actually to enact a kind of psychoanalysis with their readers. By contrast, most psychoanalytic publications in England and America engage in a dry scientific style in which each word or sign is naively assumed to have only a single—and controllable—meaning. But psychoanalysis cannot engage in an objective understanding of the child’s magical understanding. It catches the contagion of the unconscious, surreal, illogical thoughts it studies. One often feels in reading apologists for psychoanalysis that they protest too much when they proclaim its scientific status. And yet chemistry has its roots in the unreasonable formulas and ceaseless experimentation of alchemists; astronomy deceives itself if it thinks itself free of the ancient astrological need to find cosmic and mathematical correspondences between the universe and the individual.⁴ No science—and especially no writing of science—is immune from magical thinking.

Mirrorings

The phallus is the signifier “which has no signified.” In his essay “The Meaning of the Phallus,” Lacan emphasizes Freud’s discovery of an irreducible “disturbance of human sexuality.” This disturbance is “insoluble by any reduction to biological givens.”⁵ Yet Lacan explains desire—of which the

missing phallus is, from one point of view, the cause—by resorting to biological parable. Seeing itself in reflection as a unified whole, a complete body, the infant, according to Lacan, is overcome by jubilant feelings of mastery, independence, and self-control. The whole reflection contrasts sharply with the fragmentation, uncoordination, and psychic dismemberment of all the infant's previous perceptions. For Lacan, however, the crucial feature of this jubilation is its lack of foundation, its basis in illusion: the infant's perceived unity is out of step with its reality as a dependent, fractured being living in a world of perceptual non sequiturs. The baby sees itself where it is *not*: upon the reflective plane of a mirror. The child's ecstasy in the face of the mirror is the founding trick of the glorious but ultimately illusory experience of selfhood. For Lacan, the self is "decentered," it is not where it appears to be. The image of wholeness is just that: an image.

Lacan's "mirror stage" need not refer to a real glass mirror. The sense of identity and control is achieved in other ways: identification with a well-coordinated mother or other identifiably whole body. Feelings of mastery, of control, may resurge frequently in adult life, but they are always supported by the founding self-delusions of the mirror stage. Mastery, for Lacan, is always image-inary—not just "imaginary" but a special term he uses in contrast with "symbolic" and "real." It is a product of the association and identification of the mirror stage. The real I is not there or changing; it is a black letter or word, the literary equivalent of a hologram.

The mirror stage posited by Lacan relies on that peculiarity of human development discussed earlier under the name of "neoteny." Neotenous humans, remember, are those who, as adults, retained juvenile characteristics of their ape ancestors such as broad forehead, small jaws, canine teeth, and naked skin. "Actually," writes Nobel laureate François Jacob, "some of the most dramatic events in evolution resulted from a change by which sexual maturity was achieved at an earlier developmental stage, so that previously embryonic characteristics were retarded in the adult, while previously adult characteristics were lost."⁶ The mirror stage is based on this "real *specific*

prematurity of birth in man.”⁷ Many mammals walk, even run, within minutes or hours of their births, but humans, although the sense of vision quickly develops, remain helpless and hopelessly uncoordinated for more than a year. No two-year-old alone survives. The five-year-old can never feed or warm herself. Human infants are born too soon—at a stage of development when ape-relatives are safely tucked away in utero. Left alone, unable to move about or fend for itself, the human infant is nonetheless immersed in a startling display of visual imagery beyond control—a kind of psychedelic light show for a disembodied consciousness. In the mirror stage the confused infant, at the mercy of another whose breast and gaze have not yet been differentiated from itself, perceives itself in the “mirror” of the maternal body with a gleeful sense of individuality. But this glee is based on false impressions.

Moreover, the child not only receives this ecstatic, imaginary sense of being “me” from the encounter with the mirror. The encounter with the mirror also places him or her in a time frame. The baby’s frightening view of itself as being in “bits and pieces”—as in either a nightmare or a painting by Hieronymous Bosch—only occurs in retrospect, *after* the joyful encounter with an imaginary wholeness. As the American literary critic Jane Gallop writes, “What appears to precede the mirror stage is simply a projection or reflection. There is nothing on the other side of the mirror.”⁸ “What matters,” Lacan writes, “when one tries to elaborate upon some experience, isn’t so much what one understands, as what one doesn’t understand.”⁹

If Lacanian notions such as the mirror stage or the phallic signifier are found disturbing, remember that Lacan’s writings are a sort of a rebus, a picture-puzzle, and that the effort to make sense of them, to discover or invent a subtext, is *more* important than the superficial content. They are, in the parlance of literary criticism, “semiotically open texts.” From this point of view the mirror stage is a metaphor of ignorance; it is as if Lacan were saying to us, “Look, I’ll show you how this trick of the self, of identity is done: with mirrors.” But when we go to look at the mirror Lacan provides for an explanation,

we are still not quite sure if Lacan is putting us on or off—if the production of the mirror is itself a trick.

Symbols

The mirror stage is recognized by the violent disruption in the succeeding symbolic stage. The smooth “imaginary” relation between the ego and its images and between mother and child are ruptured abruptly by the fracturing and fragmenting of the father with his phallus and symbols. While similitude and identification is in the mirror stage, these break in the symbolic realm where fragmented speech and the realization of incompleteness replace the illusory wholeness of a mirror. Just as the little boy’s penis is only a phallic signifier and not the father’s penis, so the symbols and language of the symbolic realm are not the desired things themselves but only their signs.

The important identification of the psychoanalyst with the father, or the mother, or some other “significant other”—an identification that in traditional Freudian psychoanalysis is called “transference”—is part of the Lacanian image-inary. The ego—your “I”—identifies with imagoes, idealized mental images. But these images also break: the analyst is not your father, your child is not you or your younger brother. Entering Lacan’s symbolic realm we recognize the artificiality and overlap of our deepest personal identifications, and we perhaps treat others with greater awareness.

Lacan’s category of the symbolic resembles the Freudian Oedipal complex after which it is modeled. Though disruptive and fragmenting, the symbolic successor to the image stage is a necessary rite of psychological passage. Symbolization for Lacan resembles weaning; just as in the absence of the mother the rubber nipple comforts the anxious infant, the spoken words of the father provide a mixed relief. The infant begins to recognize the trick of language. Words supplement and supplant, they simultaneously designate and replace the miss-

ing objects they represent. Learning to symbolize, children cut their ties, and reconciling themselves to a chronic lack of maternal satisfaction, realize that there is no end to desire. They may identify with the speaking father—a switch by which they appropriate what once threatened them.

As words substitute for things, so the father's spoken and written law substitutes for the mother. This law is singularly prohibitive; "*le nom du père*" in Lacanian lingo literally means "the name of the father," but in spoken French the phrase sounds indistinguishable from "the 'no' of the father." The father figure by his very presence intrudes and says no to the child. And the part that stands for the whole in this troubling intrusion is the phallus, the *noeud*, the knot. For Lacan, the phallus begins the entire process of signification. A symbolic sword, severing continuous contact with the mother, it penalizes. But this phallic sword cuts both ways; by employing phallic, paternal symbolization it appropriates paternal power and the Oedipal complex is alleviated, if not resolved.

Lacan accused American psychoanalysis of betraying Freud's insights. According to Lacan, they distorted Freudian theory by deemphasizing the importance of the unconscious and of infant sexuality. American "ego analysts," by treating the Oedipus complex as a childhood phase to be outgrown in the move to a mature and healthy ego, deny the phallic signification as a continuous psychic reality. But language itself is phallic. It interrupts the primal union of the mother and child. Language, by continuously replacing and displacing what is longed for—by endlessly substituting signs—works through metaphor and metonymy. Between the lines, in the subtext—on these sheets, between these covers—is the yearning for possession of that which cannot quite be grasped. Like the Hebrew prohibition against displaying the name of God, to write anything already obviates almost all of the rest, is a betrayal, a careless slip of the writing hand. Lacan spoke of Freud's Copernican revolution as being greater than Darwin's: consciousness is eccentric, it has no center. *Ça parle*: it (the unconscious) *speaks*. At the center of human being, there is a w(hole). Disfiguring language—poeticized, perversely punc-

tuated, incomplete, or overflowing—helps reconcile us to our unavoidable “castration.” This is because such language studiously avoids the mirage of wholeness. As they say in show business, it leaves us wanting more . . .

Phallic Variety

The phallus, as we have seen, is crucial to the early magical perception of the psyche. But what of its biological cognate, the penis? How did it evolve? Both the penis and its relatively large size in humans may be the result of sperm competition. First, let's look at the human penis. In sexually very active early human societies, men with large erections (but not larger than the vaginal passage) would have gained a slight advantage in impregnating women. Inheriting their genes, the male children of these women would have been likely to grow into men with penises closer to the size of their fathers than those of ancestral apemen.

While sperm competition may adequately explain the greater relative size of human penises, evolutionists had earlier made some other suggestions: (1) that women, viewing naked males, selected as lovers those with the largest penises, (2) that males with large penises displayed them, frightening smaller-penis rivals, and (3) that females chose large-membered males because, facing frontward, these large-penis males could bring them to sexual climax. All these hypotheses are relatively unsatisfactory. Since most women, at least when queried in a woman's magazine survey, claimed to be less attracted to penis size than to other male anatomical features, the “attraction hypothesis” is easy to dismiss. The survey requiring women to rate male attributes they found sexiest reported only a 2 percent preference for penises, compared to a 39 percent interest in buttocks; the women were more attracted to broad shoulders and buttock shape than large penis size. Although comparative data on human penis size from race to race are limited, female choice of men with large penises is

apt to be most conspicuous in tropical climates. This is because women in the tropics are more apt to see males nude and judge their penises.

The “scare” hypothesis also seems unlikely: muscular men with small penises seldom cower in the presence of skinny weaklings with large ones. Unlike some other primates, human males are not known to threaten each other with erections.

Even less tempting is the hypothesis that large penises enhanced female pleasure during sexual intercourse. Penis dimension is neither the major determinant of female sexual pleasure, nor is a big penis a guarantee of female pleasure. As Stephen Jay Gould emphatically points out, the existence of an anatomical trait does not *necessarily* mean that it was an adaptation—that it conferred any benefits upon the survival of those possessing it. The sperm competition explanation for penis size is the most elegant. The vagina is, believe it or not, a very hostile place for sperm; it is acidic, a veritable obstacle course or torture chamber so full of pitfalls that most of the time not a single of the hundreds of millions of sperm ejaculated survive. A penis that reaches the back walls of the vagina therefore has an advantage over one that ejaculates its contents further from the egg. The advantage of delivering copious quantities of sperm close to the egg creates the evolutionary conditions for growth in penile length—although a penis that reaches beyond the back walls of the vagina offers no additional advantage. Human penises, unlike those of some other primates, are displayed outside the body cavity, and are therefore unusually vulnerable to injury. Our male ancestors may have had penis bones, as many mammals today do, but lost them and the fold of skin covering them in the abdominal area.

Two out of three men estimate their penis to be undersized, according to physician Barry McCarthy, author of a book on male sexual awareness.¹⁰ McCarthy attributes penis-size anxiety to several factors. First, boys catch sight of their fathers’ penises at an impressionable age and worry they won’t “catch up.” Second, glances at other males in locker rooms are made end-on: the other men’s penises seem larger because a man sees his own from above, a perspective that makes the penis

seem smaller because of the shift known to artists as foreshortening. Third, flaccid penises vary dramatically in size; erect, there is far less variation: the average human erect penis measures thirteen centimeters (five to six inches). Fourth, men are kept in ignorance by a general male reluctance to openly discuss intimate sexual matters with each other: men are more likely to tell women about penis preoccupation than they are to tell other men; and so the myth persists that large penis size enhances female pleasure.

There is much phallic variety in the biological world. Many different kinds of "penises" exist: the aedeagus of flies, mites, and butterflies, the cloacal protuberances of some frogs, the copulatory organ of the domestic honey bee drone (which suicidally breaks off and blocks all others from mating the queen), the embolus of the golden web spider, the anal fins (gonopodia) of fishes, the double hemipenes of snakes, the intromittent organs of fish and rodents, the ligula of the dragonfly. All are penises or phallic organs, used to transmit male sperm to females. Phallic variety extends from tiny protuberances to whale penises, which, though usually kept tucked away inside the body, can reach six feet in length. Ostrich males are also well endowed, so much so, in fact, that walking sticks have been fashioned from their genitalia.

The genitalia of insects are more diverse than those of mammals. The females of many insect species have "sperm storage organs," special sites that incubate sperm, keeping it alive for months prior to fertilization. Because of the remoteness of these female organs, male insects have evolved special fluted and scooplike members that bear little resemblance in form to a mammalian penis. Many times in vastly different animals unique scooping and pumping penises capable of displacing previously deposited sperm have evolved. For example, in the Zygoptera, predaceous insects in the order Odonata comprising damselflies and dragonflies, the penis has a scooplike section at the end that acts to pull clumps of competing sperm from one or both of the female's sperm storage organs. Similarly the penis bone of many mammals—the baculum—may act as a cleaning mechanism to rid the female of competing sperm.

The members of certain species of rodents and of butterflies are so ornately different in their juxtaposition of spines, curves, and fluting that as an important taxonomic tool they serve to distinguish between otherwise very similar-looking animals. The best taxonomic indicator for some species is penis shape, which evolved as males competed for access to female eggs. And there is so much variation in the size, shape, and function of damselfly penises that a qualified entomologist can identify species solely on the basis of the male organ.

The great variety of insect organs results from the mechanics of their copulation, combined with sexual selection. If it is the right size and shape, the penoid organ of another male can dislodge previously stored sperm from the sperm storage sites of insect females.

The Swedish seed bug (*Lygaeus equestris*) avoids coitus interruptus: the male has a member that not only stretches to fully two-thirds the length of his body (equivalent to a four-foot-long penis in a man!), but is replete with hooks that prevent him from being dislodged once he has engaged the female. Partner seed bugs remain joined by the male's well-placed penis for up to twenty-four hours.

Such rococo traits and macho ruses seem to have evolved as means of gaining control over sometimes reluctant female bodies; although animal sex may entail the sacrificial death of the male during the copulation, the steep price is evolutionarily worth it if it is the only way to get potentially immortal genes into the next generation. Spiky ornamentation has even developed in sperm. Snail's sperm comes adorned with backward-pointing spines that presumably act as a barbed arrowpoint to anchor one male's cells inside his female despite subsequent matings by other males.

In evolution, as in military technology, behaviors and biotechnologies occasionally appear that quickly make obsolete the highly elaborate tactics of the past. The repeater rifle, the Gatling gun, and barbed wire rendered cavalry obsolete, just as the stirrured cavalry in its turn had outmoded foot soldiery. Comparable innovations mark the twists and turns of reproductive fate in the ancient "flowery combat" of the sexes.

Consider the sexual weaponry that has evolved in the fluke *Moniliformis dubius*, a parasitic flatworm that lives in vertebrate digestive systems. These flukes plug not only the genital tracts of their females, but those of other males—as if in medieval times, chastity belt makers, enamored of their clever handiwork, had developed a model suitable for both sexes!

A most bizarre example of insect sexual weirdness is *Xylocoris maculipennis*, the African bedbug. Males possess spearlike organs with which they stab and penetrate the abdomens of females. This drastic procedure creates de facto “vaginas”—in fact, puncture wounds—at various places in female bodies. The sperm enter through such a puncture wound, swimming through the female’s hemolymph, the bloodlike body fluid. Traveling about, some reach her sperm-storage organ. There they are duly stored. The genes of the stabbers are propagated by male offspring that also possess daggerlike penises.

Such behavior must have killed ancestral female bedbugs. Yet today African bedbug females have evolved a special pad of female abdominal tissue (the “organ of Berlese”) that helps them heal. Female bedbugs possibly even use protein in the ejaculate as nutrition to help nourish their eggs. These blood-sucking denizens of third-rate hotels have also changed the rules of the mating game in another way. Using those same spearlike genitals, male African bedbugs routinely puncture other males, forcefully ejecting their sperm into the victim’s abdomen. Male–male forced matings are of course sterile; at first glance such sex seems to be gratuitous in the Darwinian game of leaving the most offspring. But bedbug sperm, unlike that of mammals, may survive for years inside the victim’s hemolymph. Merging with the victim’s sperm, the rapist’s sperm is ejaculated by proxy, through another male’s penis. And so it is that the bedbug genes, promoting what in human terms would be homosexual rape, are perpetuated into the next bedbug generation.

Most of the variation in phallic shape and function is found in insects; it is less striking in mammals. In vertebrates generally the contents of the ejaculate disperse throughout the female’s reproductive system but sperm is not stored.

Evolving Penises and Cementing Semen

Human penises did not evolve from bedbug “syringes” or other insect protrusions. The reproductive organs of men came from those of fish and later amphibian ancestors. The first penises, of which those of humans are an evolutionary elaboration, were naturally selected as they increasingly ensured fertilization of eggs. Before penises evolved, sperm was disseminated relatively randomly in water, by the fish and amphibians that were ancestral to reptiles and mammals. External fertilization—release of sperm into egg-filled water—preceded the internal fertilization in which males, equipped with penises, ejaculate sperm directly into the female body. “The penis itself might have evolved in response to sperm competition, males able to insert sperm further into the female tract and therefore closer to the ova being favored,” writes Geoff A. Parker, the innovator of sperm competition theory.¹¹

Receding into prehistory the phallus seems to disappear into the body of the evolutionary stripper. The human line extends beyond australopithecine apes to tree-dwelling primates and the first night-stalking small mammals, who themselves had evolved from the same stem reptiles that later gave rise to the dinosaurs. Beyond prosimian primates our human lineage is traced to tree-dwelling shrewlike mammals not too unlike rodents. Interestingly, males of some modern rodent species leave special mating “plugs”—gluey, sticky secretions that would bar access of other rodent males to already-inseminated females were it not that these same male rodents have also evolved special penile spines whose sole function seems to be the removal of the mating plugs. Is the peculiar stickiness of human semen, which becomes glue-like soon after ejaculation, which can mat and dry to skin and hair, a result of its ability to block future ejaculates? Drone bees leave behind both their genitalia and a sealing mucuslike substance as they mate with the queen immediately prior to their death. Compare such

mortal plugging up in bees to the obsessive guarding of females by frogs of the genus *Atelopus*; even after eggs are amply fertilized by sperm, these frogs will not let go; they continue for months to clamp on to their mates. Perhaps the viscous consistency of human semen likewise derives from our remote mammal ancestors whose semen "cemented," functioning as a kind of natural chastity belt, barring or limiting access to would-be future impregnators.

Since the first penislike organs probably appeared in amphibians transitional between water-dwelling fish and land-inhabiting reptiles, the mating patterns of existing amphibians may provide clues to penis origins. In most frogs and toads, called anurans, fertilization is external to the body; the male grasps the female from the back in a kind of long-lasting hug, called amplexus. He stays there until the female spawns, at which time he sheds sperm into the water and onto the eggs. During amplexus he may be harassed by other males attempting to push him off. This behavior of frogs and toads varies from that of the two other living orders of amphibians, the apodans and the urodeles. In both these later orders, fertilization is internal: it occurs within the body of the female. Male urodeles, including the salamanders and newts, fertilize the eggs inside the bodies of their females, but without a penis. Males instead transfer a spermatophore, a special pouch that contains bitter sperm in a sweet package of nutriment. Urodele males excrete the packages, usually leaving them as "gifts" on moist ground for the females to pick them up, eat them, and incorporate them into their bodies. (Strange as this behavior sounds to us, it has independently evolved, in almost precisely the same form, in many species of insects.) Salamander and newt amphibians display a wide variety of reproductive behaviors; the urodele group includes both water- and land-mating species. In the legless amphibians, the apodans, sperm is transferred directly into the female cloaca by means of a penislike "intromittent organ."

The fertilization that occurs in mammals inside the female body evolved from external fertilization. In the external fer-

tilization of fish such as the bluegill sunfish, the fourspine stickleback, and the blenny, eggs are released in the open water away from the body of the female; sperm fertilizes them in the open. When such fish spawn, small opportunistic males often rush into the vicinity shedding their own sperm on the eggs. Some marine animals, for example, cartilaginous fish such as sharks and rays (*Chondrichthyes*), have penislike organs, yet most male fish still release milt into the water where external fertilization occurs. With the evolution of amphibians from fish, ejaculation inside female bodies became more prevalent as males competed to get nearer and nearer to eggs. Males supplanted each other. As eggs were reached even before their release into the water, those males whose sperm reached them tended to leave more offspring. Sperm-placing intromittent organs long enough to preempt other males offered a reproductive advantage. The amphibians ancestral to reptiles and mammals most likely had such penis-precursor intromittent organs.

Another factor in the natural selection of internal fertilization and penislike organs in some species of ancestral amphibians was the harshness of the open air. Early forms of land-crawling fish and amphibians died, their sperm and eggs destroyed by solar radiation, desiccated or dispersed by winds. In organisms that returned to water for mating, fertilization and early development of the embryos fared better. Penises in other lineages may have first appeared in fish-headed amphibious ancestors, forced to survive on land as their natural aquatic habitats dried out. Although the adults live on tree bark, around the edges of ponds, on lily pads, or in the leaf litter of the forest floor, the vast majority of amphibians return to the water to breed.

Ocean cruisers, sandy beaches, and waterfalls still attract lovers. Perhaps water remains seductive because of its ancient link to animal reproduction. Males ejaculate a whitish fluid harboring haploid sperm cells that recall the four-fifths of Earth history during which life was solely microbial: moist, unprotected by hard shells or skin, protist life in the Proterozoic eon was vulnerable to the crackling heat and winds of dry land.

Many fully terrestrial animals, such as land turtles, salamanders, and tree frogs, return to water to lay their eggs. In our lineage, the fetus floats in an amniotic universe so warm and serene it is the very symbol of security and well-being. In this aquatic paradise, the fertilized egg develops, recalling the form and lifestyle of its amphibian ancestors, even to the point of resembling a tadpole. Like childbirth, human sexuality, which literally brings out the wetness of our bodies' insides, has been described as an "oceanic" experience.

Sperm-placement also arose in aquatic habitats in the absence of the penis. Male octopuses, not mammals at all but mollusks, use their tentacles to insert sperm into the orifices of female octopuses. Male octopi engaging in this act turn bright red. The example of the octopus shows that land dwelling is not a prerequisite for the evolution of fertilization within the body. Internal fertilization is more efficient, less wasteful, and, when it does evolve, tends to relegate males that do not practice it to the trash bin of male sexual evolution. The ancestors of sea horses and other teleost fish also independently evolved internal fertilization.

Once penises appeared, in many lineages they evolved to become highly ornate. Some penises may shoot sperm by propulsive ejaculation considerably further than their own length. Some lizard penises are bright blue. The sexual organs of some insects are longer than the rest of their bodies. The dogfish sprays a "spermicidal douche" that reduces a previous male's ability to deposit, displace, or dilute sperm. Penises and other sperm-placing organs vary more in shape and size than virtually any other part of animal anatomy.

Not only have male organs evolved. Female reproduction organs display a dazzling variety of patterns, too. The age-old struggle of males to oust each other to enter first is not always to the females' liking. By secluding the site in which fertilization takes place, and by evolving intricate flanges, flaps, and sphincters, the female genitals of some species are veritable obstacle courses for sperm. Both male and female reproductive systems have, of course, evolved; but since, in people, most of the female reproductive system is internal, whereas the male

reproductive system includes a penis and testicles conspicuously exposed on the outside of the body, the male genitalia play a greater role in the magical, associative thinking of the child.

Strange Bedfellows

But since the phallus is a focus of the magical thinking of the child, the evolutionary biological search for penis origins remains frustrating, if somewhat enchanted. The quest for the origin of the penis remains unresolved, in part, because fossil penis bones are not known in human ancestors, in part, because the main, fleshy part of the penis is not likely to be preserved anyway. Even in cultural artifacts, such as ancient Roman or Florentine sculptures of nudes, the penis tends to break off—often it was broken off by Christians—leaving the imagination alone to provide it.

For Lacan, psychoanalysis was a demi-science, still exciting, not yet rigidified but having to borrow its models from other sciences such as archeology and linguistics because it was still “in labor,” a new science undergoing the sociocultural difficulties of its birth. In “The Topic of the Imaginary,” Lacan exhorts his colleagues: “My dear fellows, you wouldn’t believe what you owe to geology. If it weren’t for geology, how could one end up thinking that one could move, on the same level, from a recent to a much more ancient layer? It wouldn’t be a bad thing, I note in passing, if every analyst went out and bought a small book on geology. There was once an analyst geologist, Leuba, who wrote one. I can’t recommend you to read it too highly.” In another seminar, Lacan compared his great mentor Sigmund Freud to a careful archeologist who puts back in its proper place—unlike some followers—each of the fascinating artifacts he digs up.¹² Yet after digging up the field of evolutionary biology in quest of the origins of the penis, we are left curiously unsatisfied. It is almost as if, after extensive excavation, we have returned empty-handed. Having

found the penis of evolutionary biology, we lack the phallus of psychoanalysis. If the phallus is, as psychoanalysis purports, an object so obscure it is not even clear it is an object at all, then perhaps this feeling of loss—reminiscent of waking empty-handed from a dream of glittering coins—or imagining a colorful holographic stripper from black-and-white text—is perfectly appropriate. Fundamentally different types of discourse, psychoanalysis and evolutionary biology resist merging: like the mind, which contains the body, which contains the mind, psychoanalysis and evolutionary biology each in turn attempt to engulf the other, but, like possessive lovers, neither ever fully succeeds.

Lingering Phallic Worship

Freud began his career studying the electrochemistry of nerve cells, but such empirical studies did not sustain him. His office in Vienna was stacked with hundreds of objects: archeological remains, African carvings, figurines of wise men, statuettes of mythological characters, and ornamental pieces of glass. He compared his discovery of the unconscious and the secret language of dreams to the deciphering of Egyptian hieroglyphics of the Rosetta stone. It is easily imagined that Freud's collection had a *hermas*, a limbless Greek sculpture, a carved figure with a penis but missing arms and legs. Hermas were commonly found outside dwellings in ancient Greece. The square-shaped stone pillars were surmounted by a bust or head, especially of Hermes (Hermes—associated with the Egyptian scholar Hermes Trismegistos, as well as the philosophical practice of hermaneutics—was the messenger among the gods, presiding over roads, commerce, invention, eloquence, and theft, and conducting the dead to Hades, the underworld.) The hermas or herms were part of a generalized cult in ancient Greece that included not only icons but phallic festivals and hymns of worship. In the spring of 415 B.C.,

however, vandals ravaged Athens and destroyed nearly all the hermas.

In ancient Scandinavia phallic worship was also present, as attested to by stone and bronze carvings. Ritual scenes depict men with axes, men with ploughs, men on ships; the men almost always bear both erections and swords.¹³ Even the earliest great and anonymous artwork of human beings, the cave paintings of Lascaux, depicts a falling bird-headed man with erect phallus in the presence of a speared bison. Although proud, public, overt phallic worship has declined, covert phallicism is an inalienable, if unconscious, part of the Western European–American heritage, as Lacan and Freud insist.

Named for the American president, the Hoover Tower on the campus of Stanford University is often referred to as Hoover's last erection. Was there not something phallic in Kennedy's program to send Americans to the moon? Companies and venture capitalists vie with each other to build the highest skyscrapers—the tallest towers. Bombs, rockets, and guns, undeniably of more interest to boys and men than to girls and women, also bear an unmistakable resemblance to the erect ejaculating penis.

The Sun

What then is the meaning of the phallus? For Lacan the phallus was an "object" so central to the magical psyche that its meaning was plastic, unfixed, labile. The phallus is a "signifier without a signified." It is, in other words, a pure symbol that can mean anything and therefore finally nothing.

Male baboons at the periphery of a troop have been recorded frightening members from neighboring troops by spreading their legs to grip and point their erect penises. We have already seen that, mistaking himself for a stranger, a squirrel monkey, even a youngster who has never seen another male, will similarly display his erection to the mirror in greeting. In April of 1876 the naturalist J. von Fischer wrote in *Der Zoologische*

Garten that the first time a young male mandrill of his beheld himself in a looking glass, he turned around and presented his red rear end to the mirror. Darwin read of this "mooning" and wrote to Fischer, wondering what could be the meaning of such an "indecorous habit."¹⁴ Fischer replied that he had other monkeys with equally embarrassing habits. He had taken pains, he wrote, to cure one *Macacus rhesus* of the inappropriate behavior, but the monkey continued to display its colorful rear end to strange persons and new monkeys. Fischer concluded that it must be some form of greeting. The meaning of the phallus remains as obscure as this monkey habit of displaying the backside. The phallus can be erotic arrow or indicator, a source for all pointing. Veiled in a condom, it functions transcendently, in that its sense has been removed from procreation. The phallus can be pillar, erector, cigar, hose, tool, bone, feces, limb, person. Perhaps the most innovative use of the special qualities of phallic signification to date has been accomplished by Georges Bataille, an early member of the Surrealist movement who later diverged because he felt the Surrealists were too narrow. For Bataille the entire evolutionary odyssey of humanity from low-lying bacteria to *Homo erectus* represents an erection; this erection, however, is imperfect, since the eyes of human beings are still parallel to the ground and unable to stand the sight of their true aim, the blinding sun. The sun is the most abstract and erotic object we know, suggests Bataille, since we are drawn to it but cannot look at it, since we lift ourselves toward it but never reach it.

For Bataille the odyssey of humanity would be complete if the pineal gland in the forebrain opened up and the contents of the human body came ejaculating out toward the sun. This would be the logical conclusion of phallic human evolution. Bataille's connection of the sun with sexuality is not completely ludicrous. Indeed, the sexual excitement of "spring fever" can now be provisionally traced to physiology. Corresponding with the blossoms and perfumed air are the longer days of spring; and the sun, shining later in the day, has a specific effect on humans. Exposed to light, a portion of the brain called the

superchiasmatic nucleus stimulates the pineal gland to reduce its production of melatonin, which acts as a sex entrainer or inhibitor. Arresting of melatonin production is especially sensitive to light late in the day, with the result that erotic intoxication may be a seasonally conditioned response to the brilliant sunlight and bright colors of spring. Nonetheless Bataille's idea of solar sexuality, his notion of the sun as the locus of human desire is clearly intended as a cosmic fable. A phallic fantasy.¹⁵ Far less fantastic and phallic but equally striking are the stories of cellular pleasure to be told next, in which protists cannibalized each other and bacteria engaged in rampant, if desperate, sexual practices under the radiant influence of the blazing sun.